Project: Worked examples
Title: Lifting lugs and spreader beams
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

This is a design calculation, not a thorough examination. Pin-loaded lifting lugs and the lug loads of spreader or lifting beams, checked by the BTH-1 allowable-stress approach (clause 3-3.3), by the EN 13155 proof route (2× the rated load without permanent set, 3× without release) or both: sling force from the rated load, sling angle and out-of-plane angle (with the automatic 6° tilt cases for a horizontal beam); tension beyond the hole, single-plane fracture beyond the hole, shear-out on two planes, pin bearing and dishing; capacity and utilisation per mode, the governing mode, the proof load and a marking and certification checklist. Scope notes (LOLER and PUWER): this sheet does not replace the thorough examination and inspection of lifting equipment by a competent person (LOLER reg 9), the marking, test and certification that the regulations and the standards require, the planning of the lifting operation, the weld, bolt and beam checks of the linked sheets, or the pin, shackle and sling ratings. It is available only while the lifting feature is enabled; its issue needs a competent-person attestation and a second independent check. Signatures are not authenticated by TempestOS: the signed-in names of the author, the checkers and the competent person are as entered into this installation.
Sling force: $F_v = WLL/n_{eff}$ ($n_{eff} = n_{lugs}$, at most 3); $F_s = F_v/(\cos\theta\cdot\cos\beta)$; $F_{ip} = F_s\cdot\cos\beta$; $F_{op} = F_s\cdot\sin\beta$; $N_{beam} = F_v\cdot\tan\theta$; $M_{op} = F_{op}\cdot h$ Horizontal beam, automatic 6° tilt: pitch ($\theta \pm 6°$, load redistributed, the sling at $\theta - 6°$ takes $2\cdot F_v\cdot\sin(\theta + 6°)/\sin 2\theta$); roll ($\beta + 6°$); the larger F_s governs, F_{ip}, F_{op} and N_{beam} are the envelopes Geometry: side = min(b_e, $R - D_h/2$); $b_{e,eff}$ = min(side, 4t, $0.6\cdot(f_u/f_y)\cdot\sqrt{(D_h/side)}\cdot side$); a_{eff} = a (default $R - D_h/2$); $A_t = 2\cdot t\cdot b_{e,eff}$; $A_{fr} = [1.13\cdot a_{eff} + 0.92\cdot side/(1 + side/D_h)]\cdot t$; $\varphi = 55\cdot D_p/D_h$ (degrees); $A_v = 2\cdot t\cdot[a_{eff} + (D_p/2)(1 - \cos\varphi)]$; $A_b = D_p\cdot t$ Pin clearance: $C_r = 1 - 0.275\cdot\sqrt{(1 - D_p^2/D_h^2)}$ BTH-1 (3-3.3): $F_a = \min(f_y/n_y, f_u/n_u)$; design load $D = N_d\cdot k_{svc}\cdot F_s$; $P_t = C_r\cdot F_a\cdot A_t$; $P_{sp} = C_r\cdot F_a\cdot A_{fr}$; $P_v = F_a\cdot A_v/\sqrt{3}$; $P_b = k_b\cdot F_a\cdot A_b$; $u = D/P$ for each; dishing $D_h/t \le dish_{max}$ EN 13155 (2× elastic, 3× no release): $P_{EN} = \min(Y/2, U/3)$ for each section, Y with f_y and U with f_u (shear with $f/\sqrt{3}$; fracture area without C_r); $u = F_s/P_{EN}$ Capacity: $WLL_{cap} = WLL/u_{lug}$ for the governing section of each mode (smaller of the modes used); beam: u_{beam} from N_{beam} on A_{beam}; proof load = $k_{proof}\cdot WLL$

Ref: ASME BTH-1 3-3.3 — Pin-connected plates (design of lifting lugs)
Ref: BS EN 13155 design requirements (clause to be confirmed) — Non-fixed load lifting attachments: 2× elastic, 3× no release
Ref: BS EN 13001-3-1 5 — Proof of static strength: the calculation route cited, not implemented
Ref: LOLER 1998 reg 7, reg 9 — Marking of lifting equipment; thorough examination and inspection
Ref: PUWER 1998 reg 4, reg 5 — Suitability and maintenance of work equipment

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Design mode	both		assumption: Both modes are run so the sheet shows which governs; the owner's code route decides which counts.	any
WLL	Rated load (working load limit) of the appliance, carried by all its lugs	150	kN	spec: Lifting beam specification LB-150, rated load 150 kN	0.5 … 10000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
n_lugs	Number of lugs sharing the load (a fourth is not relied on: three share)	2		drawing: LB-150-01 rev A	any
theta	Sling angle θ from the vertical, measured in the plane of the lug (the out-of-plane angle β is applied on top: F_s = F_v/(cos θ·cos β))	30	deg	drawing: LB-150-01 rev A	0 … 60 deg
beta	Out-of-plane angle β of the sling out of the lug plane	2	deg	assumption: Allowance for the sling not hanging exactly in the plane of the lug.	0 … 30 deg
beam	The lugs are on a horizontal spreader or lifting beam (adds the automatic 6° pitch and roll tilt cases)	yes		drawing: LB-150-01 rev A	any
n_cycles	Load cycles in the life of the lug (the 2×/3× route covers up to 16,000; beyond that a fatigue assessment is needed and the sheet does not run)	1000		spec: Lifting beam specification LB-150: not more than 1000 lifts in its life	1 … 16000
t	Lug plate thickness t	25	mm	drawing: LB-150-02 rev A	5 … 150 mm
Dh	Hole diameter D_h	55	mm	drawing: LB-150-02 rev A	10 … 500 mm
Dp	Pin diameter D_p (not larger than the hole)	50	mm	drawing: LB-150-02 rev A	8 … 500 mm
R	Lug end radius R about the hole centre (the plate is 2R wide at the hole)	75	mm	drawing: LB-150-02 rev A	15 … 600 mm
be	Edge distance b_e from the hole edge to the plate edge on each side, across the load line (limited to R − D_h/2; the effective width for tension is reduced further by 4t and f_u/f_y)	40	mm	drawing: LB-150-02 rev A	5 … 400 mm
h	Distance h from the hole centre to the weld or base, for the out-of-plane moment (blank: no moment given)	180	mm	drawing: LB-150-02 rev A	1 … 3000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_y	Yield strength f_y of the lug plate for its thickness	345	MPa	spec: S355 plate 25 mm, mill certificate MC-2291 (16 mm < t ≤ 40 mm)	100 … 1100 MPa
f_u	Ultimate strength f_u of the lug plate	470	MPa	spec: S355 plate 25 mm, mill certificate MC-2291 (lower bound of the ultimate range)	150 … 1400 MPa
Nd	Design factor N_d of the Design Category (BTH-1 mode; user record)	1.8		assumption: Illustrative user-record value for this worked example, not a value from any standard: replace with the design factor of your Design Category from your own copy of the code.	1 … 5
n_y	Divisor n_y on f_y; the allowable stress is F_a = min(f_y/n_y, f_u/n_u) (BTH-1 mode; user record)	1.6		assumption: Illustrative user-record value, not from a standard: replace with your own record.	1 … 4
n_u	Divisor n_u on f_u; the allowable stress is F_a = min(f_y/n_y, f_u/n_u) (BTH-1 mode; user record)	2.2		assumption: Illustrative user-record value, not from a standard: replace with your own record.	1 … 6
k_b	Bearing multiple k_b of the allowable stress on the projected pin area (BTH-1 mode; user record)	1.2		assumption: Illustrative user-record value, not from a standard: replace with your own record.	0.5 … 3
dish_max	Dishing-ratio limit on D_h/t (blank: the dishing check is not made; user record)	40		assumption: Illustrative user-record value, not from a standard: replace with your own record.	1 … 100

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_proof	Proof-load factor on the rated load (blank: no proof load given; user record)	1.5		assumption: Illustrative user-record value, not from a standard: replace with the proof-load factor of your procedure.	1 ... 3
A_beam	Gross cross-section area of the spreader or lifting beam, for its axial utilisation (blank: lug utilisation only)	80	cm^2	drawing: LB-150-03 rev A	1 ... 50000 cm^2

3. Assumptions

A1. A design calculation, not a thorough examination: it checks the lug plate at the pin hole against the user's chosen criteria and nothing else. (LOLER reg 9 and PUWER duties are the owner's and the competent person's.)
A2. Nominal-stress screening of four sections (tension beyond the hole, single-plane fracture beyond the hole, shear-out, bearing) on a plate loaded through a pin; no stress-concentration factor, no fatigue (the cycle count is limited to 16,000), no weld, bolt or beam check, which are made on the linked sheets. (The notch-stress route of EN 13001-3-1 is cited, not implemented.)
A3. The load on the hole is the resultant sling force F_s whatever the out-of-plane angle, which is conservative; the out-of-plane component and its moment are outputs for the weld and base checks. (F_op = F_s·sin β.)
A4. Equal sharing between the lugs, with a fourth lug not relied on (three share). The share of a statically indeterminate set is not computed. (Conservative convention; replace with the actual share if it is known.)
A5. A horizontal beam adds the automatic 6° tilt as two cases: pitch (sling angles θ ± 6° with the load redistributed between the lugs by statics, the slings meeting at the hook) and roll (out-of-plane angle + 6°), the larger sling force governing. With θ ≤ 6° the pitch case puts the whole load on one lug; with more than two lugs the two-lug redistribution is applied to each lug's share. (Addendum B §B6 S8; EN 13155.)
A6. The allowable stress is the one lesser value min(f_y/n_y, f_u/n_u) for every BTH-1 section, as in the published worked example the sheet is checked against; the pin-clearance factor C_r is applied as stated from D_p/D_h < 1 with no cut-off. (Conservative; owner checklist.)
A7. Signatures are not authenticated by TempestOS: the platform asks for no password, so the author, the two checkers and the competent person are as entered into this installation; any person can switch to another name. (ADR-0173; the feature stays off until this is addressed.)
A8. The design factor, service-class factor, allowable-stress divisors, bearing multiple, dishing limit and proof-load factor are the user's records from their own licensed copy of the code; none is shipped and none has a default. (ASME BTH-1 and EN 13155 text and tables are not shipped.)
A9. The equations are written out in the spec document in the author's notation (the single-plane fracture and effective-width relations and the 1.13 and 0.92 constants follow the published course example); the owner compares them with BTH-1 (2023 or later) and EN 13155+A1 before the feature is ever enabled. (Owner test pack, S8 review checklist.)
A10. mode (Design mode) is taken as both. (Both modes are run so the sheet shows which governs; the owner's code route decides which counts.)
A11. beta (Out-of-plane angle β of the sling out of the lug plane) is taken as 2 deg. (Allowance for the sling not hanging exactly in the plane of the lug.)
A12. Nd (Design factor N_d of the Design Category (BTH-1 mode; user record)) is taken as 1.8. (Illustrative user-record value for this worked example, not a value from any standard: replace with the design factor of your Design Category from your own copy of the code.)
A13. n_y (Divisor n_y on f_y; the allowable stress is F_a = min(f_y/n_y, f_u/n_u) (BTH-1 mode; user record))

is taken as 1.6. (Illustrative user-record value, not from a standard: replace with your own record.)
A14. n_u (Divisor n_u on f_u; the allowable stress is F_a = min(f_y/n_y, f_u/n_u) (BTH-1 mode; user record)) is taken as 2.2. (Illustrative user-record value, not from a standard: replace with your own record.)
A15. k_b (Bearing multiple k_b of the allowable stress on the projected pin area (BTH-1 mode; user record)) is taken as 1.2. (Illustrative user-record value, not from a standard: replace with your own record.)
A16. dish_max (Dishing-ratio limit on D_h/t (blank: the dishing check is not made; user record)) is taken as 40. (Illustrative user-record value, not from a standard: replace with your own record.)
A17. k_proof (Proof-load factor on the rated load (blank: no proof load given; user record)) is taken as 1.5. (Illustrative user-record value, not from a standard: replace with the proof-load factor of your procedure.)

4. Working

[19] n_eff = min(n_lugs, 3) spec
= min(2, 3)
= 2
Lugs sharing the load (a fourth is not relied on)

[20] F_v = WLL/n_eff spec
= 150 kN / 2
= 75 kN
Vertical load on each lug

[21] F_s_nom = F_v/(cos □·cos □) spec
= 75 kN / (cos 30° × cos 2°)
= 86.655 kN
Sling force on each lug, untilted

[22] F_s_pitch = 2·F_v·sin(□ + 6°)/sin 2□ / cos □ EN 13155
= 2 × 75 kN × sin(36°) / sin(60°) / cos(2°)
= 101.87 kN
Sling force on the more loaded lug when the beam pitches 6°

[23] F_s_roll = F_v/(cos □·cos(□ + 6°)) EN 13155
= 75 kN / (cos 30° × cos 8°)
= 87.454 kN
Sling force when the beam rolls 6°

[24] F_s = max(F_s,nom, F_s,pitch, F_s,roll) spec
= governing case: pitch
= 101.87 kN
Sling force on each lug (resultant), governing case

[25] F_ip = max over the cases of F_s·cos □ spec
= 101.87 kN × cos □
= 101.81 kN
In-plane component of the sling force (envelope)

[26] F_op = max over the cases of F_s·sin □ spec
= 101.87 kN × sin □
= 12.171 kN
Out-of-plane component of the sling force (envelope)

[27] N_beam = max over the cases of the horizontal component spec
= 75 kN × tan(30°)
= 43.301 kN
Horizontal sling component per lug (axial force in a spreader beam, envelope)

[29] M_op = F_op·h spec
= 12.171 kN × 180 mm
= 2.1908 kN·m
Out-of-plane moment at the weld or base

[30] side = min(b_e, R − D_h/2) spec
= min(40, 75 − 55/2) mm
= 40 mm

Edge distance on each side of the hole

[31] a_eff = a, or R − D_h/2 — spec

= 75 − 55/2 mm

= 47.5 mm

End distance along the load line

[32] phi = 55·D_p/D_h (degrees) — BTH-13-3.3

= 55 × 50 / 55

= 50 deg

Angle of the shear planes

[33] be_eff = min(b_e,eff, 4t, 0.6·(f_u/f_y)·√(D_h/b_e,eff)·b_e,eff) — BTH-13-3.3

= min(40, 100, 38.339) mm

= 38.339 mm

Effective width beyond the hole in tension

[34] A_t = 2·t·b_e,eff — BTH-13-3.3

= 2 × 25 × 38.339 mm

= 1917 mm²

Area beyond the hole in tension

[35] A_fr = (1.13·a + 0.92·b_e/(1 + b_e/D_h))·t — BTH-13-3.3

= (1.13 × 47.5 + 0.92 × 40 / (1 + 40/55)) × 25 mm

= 1874.5 mm²

Single-plane fracture area beyond the hole (before C_r)

[36] A_v = 2·t·[a + (D_p/2)(1 − cos □)] — BTH-13-3.3

= 2 × 25 × [47.5 + (50/2) × (1 − cos 50°)] mm

= 2821.5 mm²

Shear area of the two shear-out planes

[37] A_b = D_p·t — spec

= 50 × 25 mm

= 1250 mm²

Projected bearing area of the pin

[38] Cr = 1 − 0.275·√(1 − D_p²/D_h²) — BTH-13-3.3

= 1 − 0.275 × √(1 − (50/55)²)

= 0.88544

Pin-clearance factor

[39] Fa = min(f_y/n_y, f_u/n_u) — spec

= min(345 / 1.6, 470 / 2.2) MPa

= 213.64 MPa

Allowable stress

[40] D_bth = N_d·k_svc·F_s — BTH-13-3.3

= 1.8 × 1 × 101.87 kN

= 183.37 kN

Design load on each lug

[41] P_t = C_r·F_a·A_t — BTH-13-3.3

= 0.88544 × 213.64 MPa × 1917 mm²

= 362.61 kN

Tension beyond the hole

[42] P_sp = C_r·F_a·A_fr — BTH-13-3.3

= 0.88544 × 213.64 MPa × 1874.5 mm²

= 354.58 kN

Single-plane fracture beyond the hole

[43] P_v = F_a·A_v/√3 — BTH-13-3.3

= 213.64 MPa × 2821.5 mm² / √3

= 348.01 kN

Double-plane shear-out

[44] $P_b = k_b \cdot F_a \cdot A_b$ BTH-1 3-3.3
= 1.2 × 213.64 MPa × 1250 mm²
= 320.45 kN
Pin bearing
[45] $u_t = D/P$ BTH-1 3-3.3
= 183.37 kN / 362.61 kN
= 0.50568
Utilisation, tension beyond the hole
[46] $u_{sp} = D/P$ BTH-1 3-3.3
= 183.37 kN / 354.58 kN
= 0.51713
Utilisation, single-plane fracture
[47] $u_v = D/P$ BTH-1 3-3.3
= 183.37 kN / 348.01 kN
= 0.52689
Utilisation, double-plane shear
[48] $u_b = D/P$ BTH-1 3-3.3
= 183.37 kN / 320.45 kN
= 0.5722
Utilisation, bearing
[49] $u_{bth} = \max(u_t, u_{sp}, u_v, u_b)$ BTH-1 3-3.3
= governing BTH-1 section
= 0.5722
BTH-1 utilisation
[50] $cap_{bth} = WLL/u_{bth}$ BTH-1 3-3.3
= 150 kN / 0.5722
= 262.14 kN
Rated load the lug carries by the BTH-1 approach
[52] $dish = D_h/t$ spec
= 55 / 25
= 2.2
Dishing ratio
[53] $u_{dish} = (D_h/t)/dish_{max}$ spec
= 2.2 / 40
= 0.055
Dishing-ratio utilisation
[54] $P_{en_t} = \min(Y/2, U/3)$ EN 13155
= min(661.35 kN / 2, 900.97 kN / 3)
= 300.32 kN
EN 13155 capacity, tension beyond the hole
[55] $u_{en_t} = F_s/P_{EN}$ EN 13155
= 101.87 kN / 300.32 kN
= 0.3392
EN 13155 utilisation, tension beyond the hole
[56] $P_{en_sp} = \min(Y/2, U/3)$ EN 13155
= min(646.7 kN / 2, 881.02 kN / 3)
= 293.67 kN
EN 13155 capacity, single-plane fracture
[57] $u_{en_sp} = F_s/P_{EN}$ EN 13155
= 101.87 kN / 293.67 kN
= 0.34688
EN 13155 utilisation, single-plane fracture

[58] P_en_v = min(Y/2, U/3) EN 13155
= min(562.01 kN / 2, 765.63 kN / 3)
= 255.21 kN
EN 13155 capacity, double-plane shear
[59] u_en_v = F_s/P_EN EN 13155
= 101.87 kN / 255.21 kN
= 0.39916
EN 13155 utilisation, double-plane shear
[60] P_en_b = min(Y/2, U/3) EN 13155
= min(431.25 kN / 2, 587.5 kN / 3)
= 195.83 kN
EN 13155 capacity, bearing
[61] u_en_b = F_s/P_EN EN 13155
= 101.87 kN / 195.83 kN
= 0.52018
EN 13155 utilisation, bearing
[62] u_en = max over the four sections EN 13155
= governing EN 13155 section
= 0.52018
EN 13155 utilisation
[63] cap_en = WLL/u_en EN 13155
= 150 kN / 0.52018
= 288.36 kN
Rated load the lug carries by the EN 13155 route
[64] lug_util = max over the modes used spec
= governing lug section
= 0.5722
Governing lug utilisation
[65] capacity = WLL/lug_util spec
= 150 kN / 0.5722
= 262.14 kN
Rated load the lug carries in the governing mode
[67] u_beam = max(N_d·k_svc·N_beam/(A·F_a), max(2N/(A·f_y), 3N/(A·f_u))) spec
= N_beam = 43.301 kN on 80 cm²
= 0.045605
Axial yield utilisation of the beam (buckling: S7)
[68] util = max(lug_util, u_beam, u_dish) spec
= max(0.5722, 0.045605, 0.055)
= 0.5722
Governing utilisation of the lug and the spreader
[70] proof_load = k_proof·WLL spec
= 1.5 × 150 kN
= 225 kN
Proof-load value

5. Results

SYMBOL	VALUE	DESCRIPTION
governing_mode	"BTH-1: bearing"	Governing mode and section
capacity	262.14 kN	Rated load the lug carries in the governing mode
util	0.5722	Governing utilisation of the lug and the spreader
lug_util	0.5722	Governing lug utilisation
cap_bth	262.14 kN	Capacity, BTH-1 mode

SYMBOL	VALUE	DESCRIPTION
u_bth	0.5722	Utilisation, BTH-1 mode
cap_en	288.36 kN	Capacity, EN 13155 mode
u_en	0.52018	Utilisation, EN 13155 mode
u_beam	0.045605	Beam axial utilisation
F_s	101.87 kN	Sling force on each lug
F_ip	101.81 kN	In-plane component
F_op	12.171 kN	Out-of-plane component
N_beam	43.301 kN	Horizontal component per lug
M_op	2.1908 kN·m	Out-of-plane moment
F_s_pitch	101.87 kN	Sling force, beam pitched 6°
F_s_roll	87.454 kN	Sling force, beam rolled 6°
side	40 mm	Edge distance on each side of the hole
be_eff	38.339 mm	Effective width beyond the hole
a_eff	47.5 mm	End distance along the load line
Cr	0.88544	Pin-clearance factor C_r
phi	50 deg	Shear-plane angle φ
A_v	2821.5 mm²	Shear area
A_t	1917 mm²	Tension area beyond the hole
A_fr	1874.5 mm²	Fracture area beyond the hole
A_b	1250 mm²	Bearing area
Fa	213.64 MPa	Allowable stress
D_bth	183.37 kN	Design load on each lug
P_t	362.61 kN	Tension beyond the hole
P_sp	354.58 kN	Single-plane fracture
P_v	348.01 kN	Double-plane shear
P_b	320.45 kN	Bearing
u_t	0.50568	Utilisation, tension
u_sp	0.51713	Utilisation, fracture
u_v	0.52689	Utilisation, shear
u_b	0.5722	Utilisation, bearing
dish	2.2	Dishing ratio
u_dish	0.055	Dishing utilisation
P_en_t	300.32 kN	EN 13155 capacity, section t
u_en_t	0.3392	EN 13155 utilisation, section t
P_en_sp	293.67 kN	EN 13155 capacity, section sp

SYMBOL	VALUE	DESCRIPTION
u_en_sp	0.34688	EN 13155 utilisation, section sp
P_en_v	255.21 kN	EN 13155 capacity, section v
u_en_v	0.39916	EN 13155 utilisation, section v
P_en_b	195.83 kN	EN 13155 capacity, section b
u_en_b	0.52018	EN 13155 utilisation, section b
proof_load	225 kN	Proof-load value
sheet_notice	"This is a design calculation, not a thorough examination. It does not replace the thorough examination by a competent person (LOLER reg 9) or the marking, testing and certification that the regulations require. Signatures are not authenticated by TempestOS: the signed-in names of the author, the checkers and the competent person are as entered into this installation."	Status of this sheet
checklist	"(1) mark the appliance with its rated load and a unique identification (LOLER reg 7); (2) thorough examination by a competent person before first use and at the intervals required (LOLER reg 9) — this sheet is not one; (3) proof load 225 kN, test certificate where the route requires it; (4) material certificates for the lug plate and the pin, and weld procedure and inspection records for the lug …	Marking and certification checklist

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	BTH-1: bearing	u_b = 0.5722	u_b ≤ 1	0.572	BTH-1 3-3.3
PASS	BTH-1: double-plane shear	u_v = 0.52689	u_v ≤ 1	0.527	BTH-1 3-3.3

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	EN 13155 (3×WLL no release): bearing	u_en_b = 0.52018	u_en_b ≤ 1	0.52	EN 13155
PASS	BTH-1: single-plane fracture	u_sp = 0.51713	u_sp ≤ 1	0.517	BTH-1 3-3.3
PASS	BTH-1: tension beyond the hole	u_t = 0.50568	u_t ≤ 1	0.506	BTH-1 3-3.3
PASS	EN 13155 (3×WLL no release): double-plane shear	u_en_v = 0.39916	u_en_v ≤ 1	0.399	EN 13155
PASS	EN 13155 (3×WLL no release): single-plane fracture	u_en_sp = 0.34688	u_en_sp ≤ 1	0.347	EN 13155
PASS	EN 13155 (3×WLL no release): tension beyond the hole	u_en_t = 0.3392	u_en_t ≤ 1	0.339	EN 13155
PASS	BTH-1: dishing ratio	u_dish = 0.055	u_dish ≤ 1	0.055	spec
PASS	Spreader beam axial yield	u_beam = 0.045605	u_beam ≤ 1	0.0456	spec

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: f_y, f_u (Material properties); Nd (Design factor and service-class factor); n_y, n_u, k_b (Allowable-stress divisors and bearing multiple); dish_max, k_proof (Dishing limit and proof-load factor). Enter each as one so its source, edition and licence travel with it.
WARNING — k-svc: k_svc: no service-class factor entered, 1.0 used; enter it from your own copy of the code if it applies.
WARNING — tilt: beam: the automatic 6° tilt cases are applied (pitch: sling angles θ ± 6° with the load redistributed between the lugs; roll: out-of-plane angle + 6°) and the larger sling force governs.
WARNING — be-eff: be: the effective width 38.339 mm is below the edge distance 40 mm (limited to 4t and by f_u/f_y), so the tension area uses the smaller.

8. Verification

```
Template struct.lifting@1.1.0 Content hash b2c8e9b2c9a962c1 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```